Lab: Case Study Analysis — Neural Self-Organization

Objective

Analyze clinical and experimental cases that illustrate the brain as a self-organizing system, applying the Active Inference framework to interpret neural data.

Part 1: Resting-State fMRI Analysis

Goal: Interpret resting-state neural data through the generative model lens.

Case: A healthy 30-year-old participant undergoes a 10-minute resting-state fMRI scan. Independent component analysis reveals three major networks: the default mode network (DMN), salience network, and central executive network. The DMN shows highest activity during rest; the salience network activates during stimulus detection; the executive network activates during cognitive tasks.

- Map each network to a proposed component of the brain's generative model (self-model, precision weighting, policy selection).
- What would Active Inference predict about the *temporal dynamics* of switching between these networks?
- Design a follow-up experiment that could test this prediction.

Write your response here

Part 2: Anosognosia After Stroke

Goal: Analyze a clinical case of failed model updating.

Case: Patient M.R., 67 years old, suffered a right parietal stroke resulting in left hemiplegia (inability to move the left arm). Despite clear evidence of paralysis, M.R. insists her arm is functioning normally. When asked to lift both arms, she lifts only her right arm but reports that both are raised.

- Interpret anosognosia through Active Inference: what generative model component has failed to update?
- Why might right parietal damage specifically cause this deficit? (Hint: the right hemisphere's role in body schema and precision on self-model prediction errors.)
- How does this case illustrate the claim that "experience is shaped by the generative model, not just by sensory input"?

Write your response here

Part 3: Sensory Deprivation and Hallucination

Goal: Explore what happens when the generative model runs without sensory constraint.

Case: Participants in a sensory deprivation study (Ganzfeld experiment) report visual hallucinations after approximately 20 minutes of uniform visual input. The hallucinations range from simple geometric patterns to complex formed images.

- How does Active Inference explain the emergence of hallucinations under sensory deprivation?
- What does this tell us about the relationship between "bottom-up" sensory input and "top-down" generative model predictions?
- What would you predict about the content of hallucinations in participants with different expertise (e.g., musicians vs. visual artists)?

Write your response here

Part 4: Neural Correlates of Prediction Error

Goal: Map the theoretical concept to empirical measurements.

Case: An EEG study measures the mismatch negativity (MMN) response — a negative voltage deflection occurring when an unexpected auditory stimulus violates a regular pattern (e.g., a deviant tone in a series of standard tones). The MMN peaks at approximately 150-250 ms after stimulus onset.

- Interpret the MMN as a neural signature of prediction error. What levels of the generative model are involved?
- Why does the MMN amplitude decrease with repeated deviants (habituation)? How does Active Inference explain this in terms of model updating?
- The MMN is reduced in schizophrenia. What does this suggest about the generative model in schizophrenia (hint: precision on prediction errors)?

Write your response here

Part 5: Synthesis

Goal: Integrate the cases into a unified view of the brain as a self-organizing system.

Write a 200-word synthesis connecting the four cases above: how do they collectively illustrate the brain as a system that maintains a generative model, generates prediction errors, and updates through inference?

Write your response here

Lab Summary

Part	Skill Developed	Case Type
1	Network interpretation	Resting-state fMRI data analysis
2	Clinical reasoning	Neurological deficit (anosognosia)
3	Experimental prediction	Sensory deprivation and hallucination
4	Electrophysiology analysis	Mismatch negativity and prediction error
5	Integrative synthesis	Connecting four cases to the systems framework